Company: Equinor | Well: 31/5-7 | Field: Eos | Rig Name: West Hercules | Country: Norway

Drilling Mechanics - APWD Time Based
Real Time, Recorded Mode

Schlumberger

Company:	Equinor
Well:	31/5-7
Field:	Eos
Rig Name:	West Hercules
Country:	Norway

Latitude:	60° 34' 35.14" N	UWID:	
Longitude:	3° 26' 36.13" E	Rig Name:	West Hercules
Block:	31/5	Rig Type:	Semi-Submersible

FL: Norwegian North Sea
FL1: Section: 8 1/2 in. pilot
FL2: Job no.: 19SCA0085

Log Measured From: - Drill Floor: 31.00 m
Permanent Datum: - Mean Sea Level
Ground Level: 307.00 m

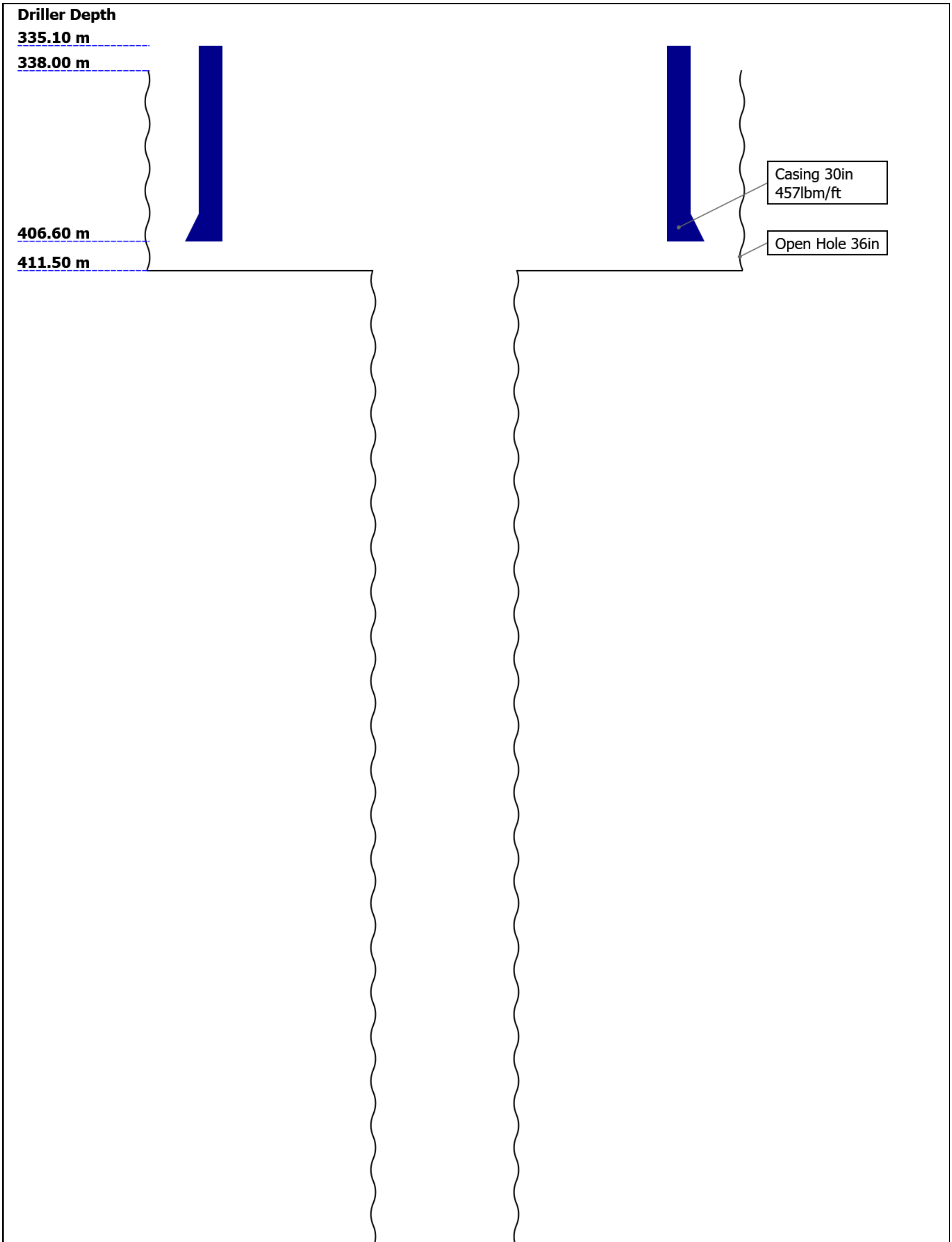
Acquisition Dates:	04-Dec-2019
Log Interval:	12/4/2019 1:00:00 AM -- 12/4/2019 8:15:00 PM
Index Types:	Time
Index Scales:	5 cm / 3600 secs
Depth Source:	Driller's Depth
Depth Sensor:	3rd Party Depth
Print Type:	Final
Spud Date:	02-Dec-2019

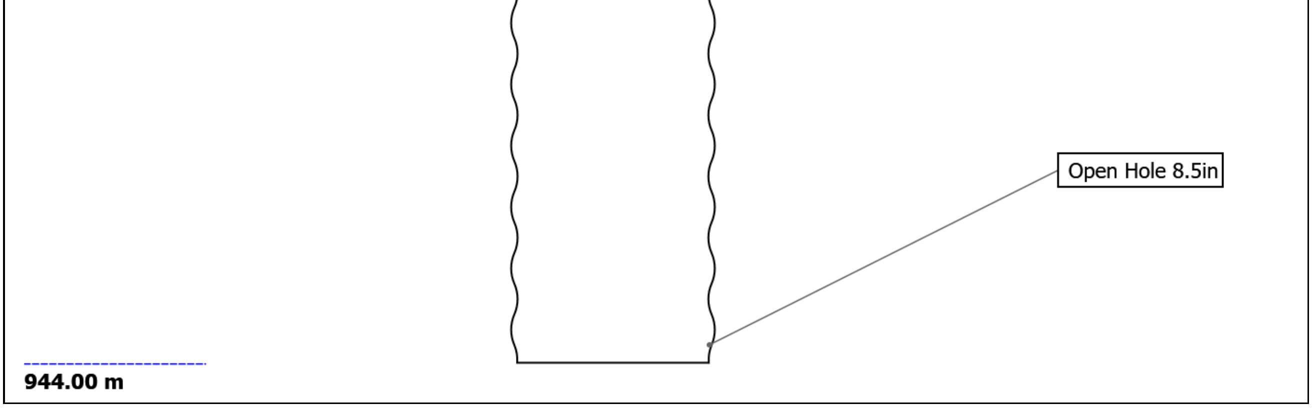
Other Services:
Direction & Inclination (Surveys)
Directional Drilling
EcoScope Service
SonicScope

Disclaimer

Contents

Well Sketch


Driller Depth
335.10 m
338.00 m
406.60 m
411.50 m
Casing 30in
457lbm/ft
Open Hole 36in



Borehole Size/Casing Record

Bit							
Bit Size (in)	36	8.5					
Top Driller (m)	338	411.5					
Bottom Driller (m)	411.5	944					
Casing							
Size (in)	30						
Weight (lbm/ft)	457						
Inner Diameter (in)	27.067						
Grade	X56						
Top Driller (m)	335.1						
Bottom Driller (m)	406.6						

Operational Run Summary

Parameter (unit)	**Run 2**						
Date Log Started	04-Dec-2019						
Time Log Started	00:53:56						
Date Log Finished	04-Dec-2019						
Time Log Finished	20:23:38						
Bit Size (in)	8.500						
Bit Start Depth (m)	411.50						
Bit Stop Depth (m)	944.00						
Top Log Interval (m)	406.80						
Bottom Log Interval (m)	936.20						
Max Hole Deviation (deg)	0.18						
Azimuth of Max Deviation (deg)	108.58						
Logging Unit Number	A615						
Logging Unit Location	Service Office						
Recorded By	J.Johansen						
Witnessed By	R.Nordstokke						
Service Order Number	19SCA0085						

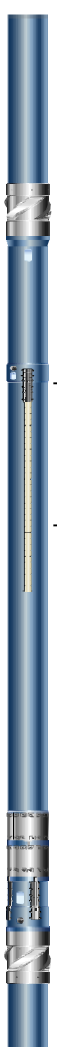
Borehole Fluids

Parameter(unit)	Run 2						
Fluid Type	Water						
Fluid Name	Seawater						
Max Recorded Temperatures (degC)	16						
Source of Sample	Active Tank						
Salinity (ppm)	33579.09						
Density (g/cm3)	1.03						
Funnel Viscosity (s)							
Fluid Loss (cm3)							
PH							
Source RMF							
RMC	Pressed						
RM @ Meas Temp (ohm.m@degC)	0.19 @ 23.9						
RMF @ Meas Temp (ohm.m@degC)	0.19 @ 23.9						
RMC @ Meas Temp (ohm.m@degC)							
RM @ BHT (ohm.m@degC)	0.23 @ 16						
RMF @ BHT (ohm.m@degC)	0.23 @ 16						
RMC @ BHT (ohm.m@degC)	NaN @ 16						
Total Solid (%)							
High Gravity Solids (%)							

Remarks and Equipment Summary

Run 2: Toolstring

Equip name	Length	MP name	Offset
SONICSCOPE6:H0121	**32.32**	SonicScope	
MP6DC:H0121			
Upper Stabilizer:00164-01			
Lower Stabilizer:00370-2-2			
Upper Extender			
MP6UC:8019			
MP6LC:8019			
Lower Extender:RFE-11			
XOS6:78001			
		ROP	29.33
		Delta-T	28.12



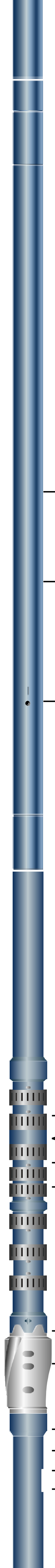
Run 2: Remarks

All depths are referenced to driller's depth. See EOWR for depth tracking details.

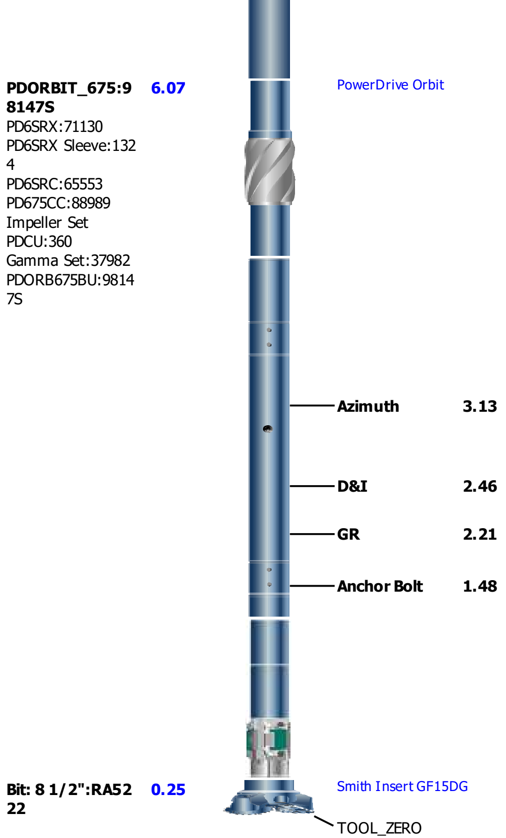
APWD data from tool memory. All other data is Real Time.

Gamma Ray measurement is environmentally corrected for mud weight, bit size, collar thickness and neutron activation.

Run 2 and 8 1/2 in. pilot section TD at 944 m.



TELE675:E4699
22.34
MSSU675:12775-9
Upper Extender:08-118
MDC675:E4699
MMA:3766
MDI:1277
PMVC:560
PMEA:3491
MTA:2375
MTK675:1510/298915
MSSD675:916772-1
Lower Extender, Long:319
TeleScope
D&I 18.11
Vibration 17.11
ROP 15.76
DV6MT:2855
13.96
DVMC:2855
DVMS:3811
UpperExtender:1954
PNG:8718-51675
DVME:1801
GGLS:A2609
LowerExtender:08/044
MSSA:83777
EcoScope
PNG Monitor 11.56
Sigma 11.36
Spectroscopy 11.36
Neutron 11.06
Resistivity 10.81
UltraSonic 9.39
Density 8.97
Inclinometer 8.29
ROP 8.06
Pressure 7.93
GR 7.77



Lengths are in m
Maximum Outer Diameter = 8.500 in
Line: Sensor Location, Value: Gating Offset
All measurements are relative to TOOL_ZERO

Run 2

Drilling Mechanics - APWD - Time Based

Pass Summary

Run Name	Pass Objective	Direction	Start	Stop	Include Parallel Data
Run 2	TimeLogicalAcq	Down	04-Dec-2019 12:53:56 AM	04-Dec-2019 8:23:38 PM	Yes

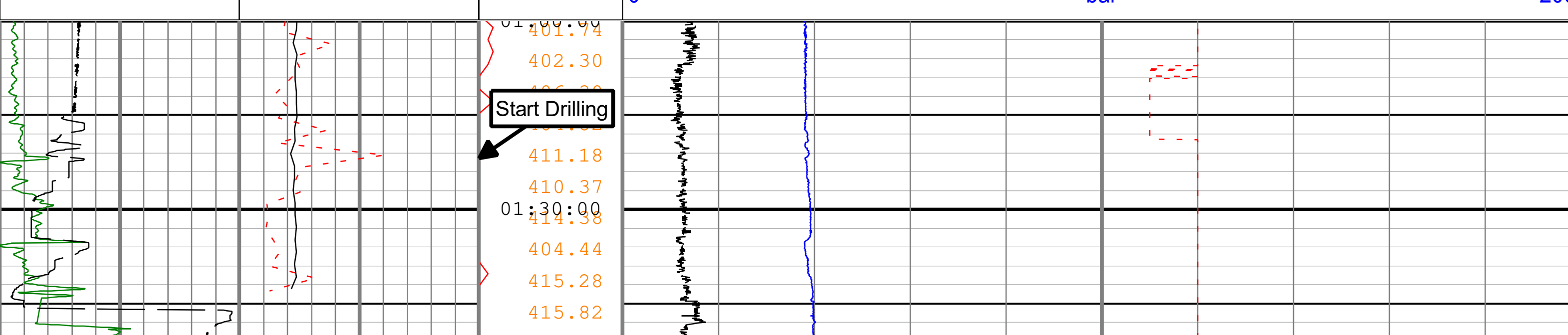
Log

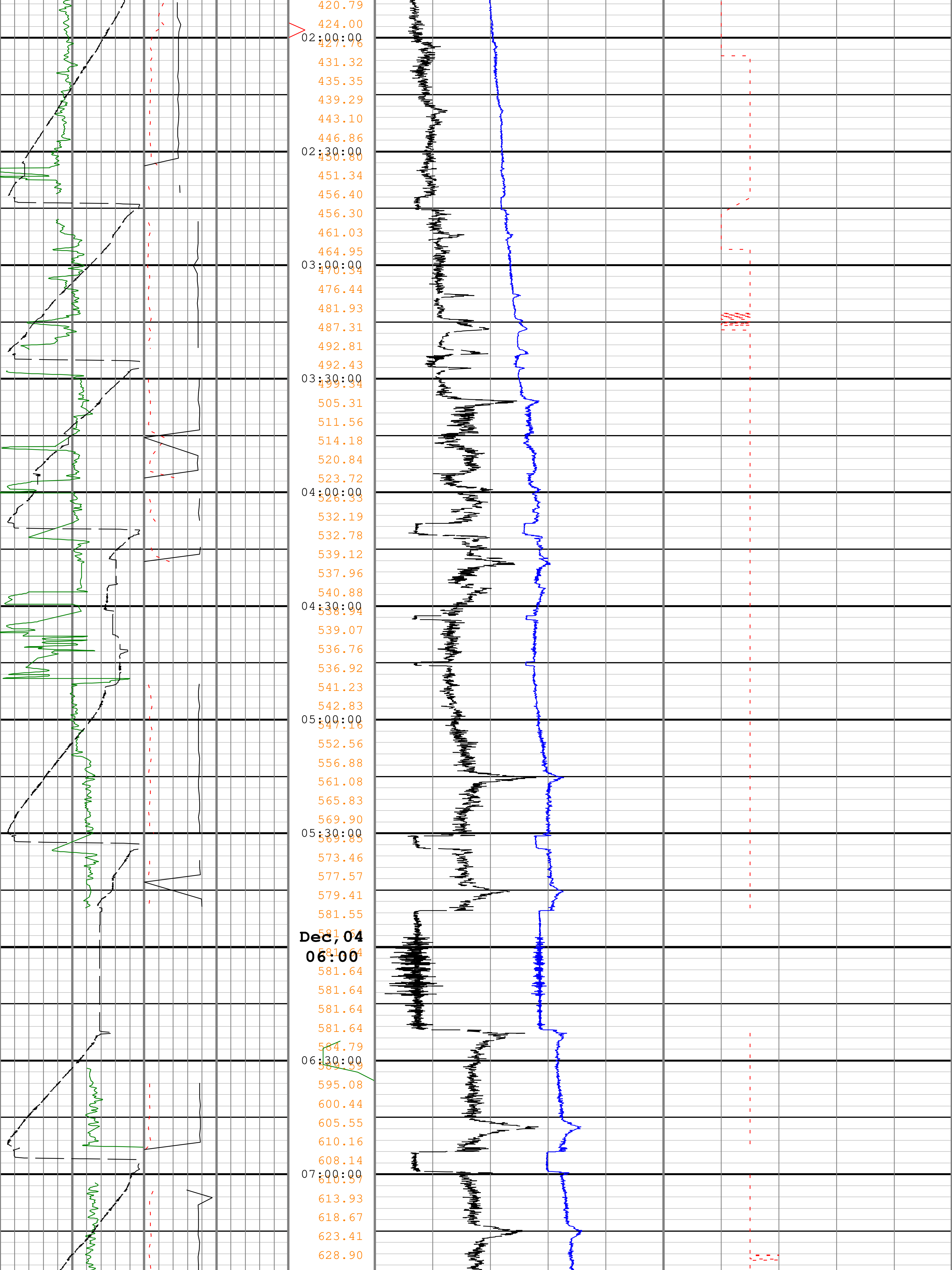
Company:Equinor Well:31/5-7

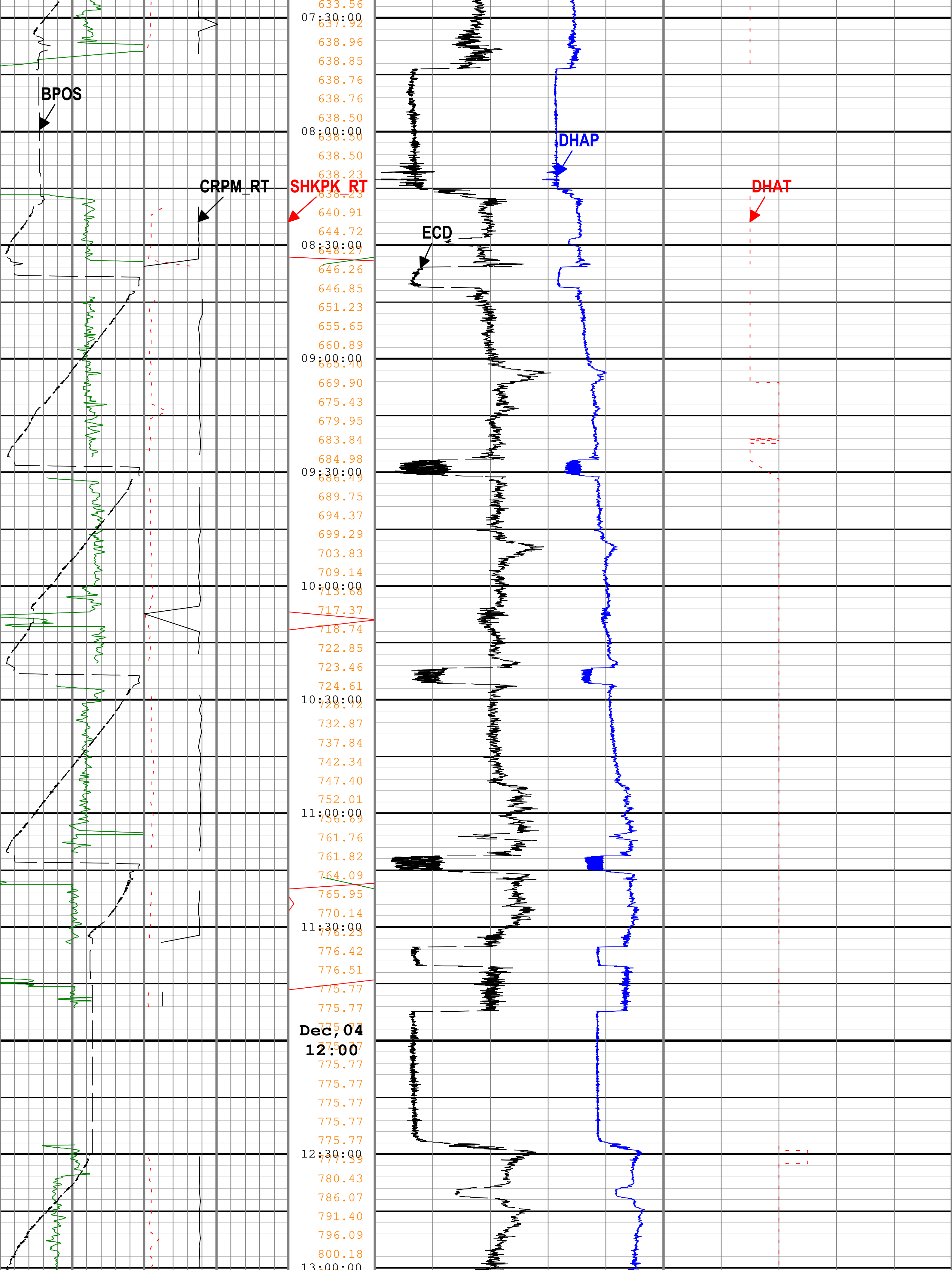
Run 2: TimeLogicalAcq:S026

Description: Format: Log (Drilling Mechanics Light) Index Scale: 5 cm per 3600 s Index Type: Time Creation Date: 11-Dec-2019 15:17:16

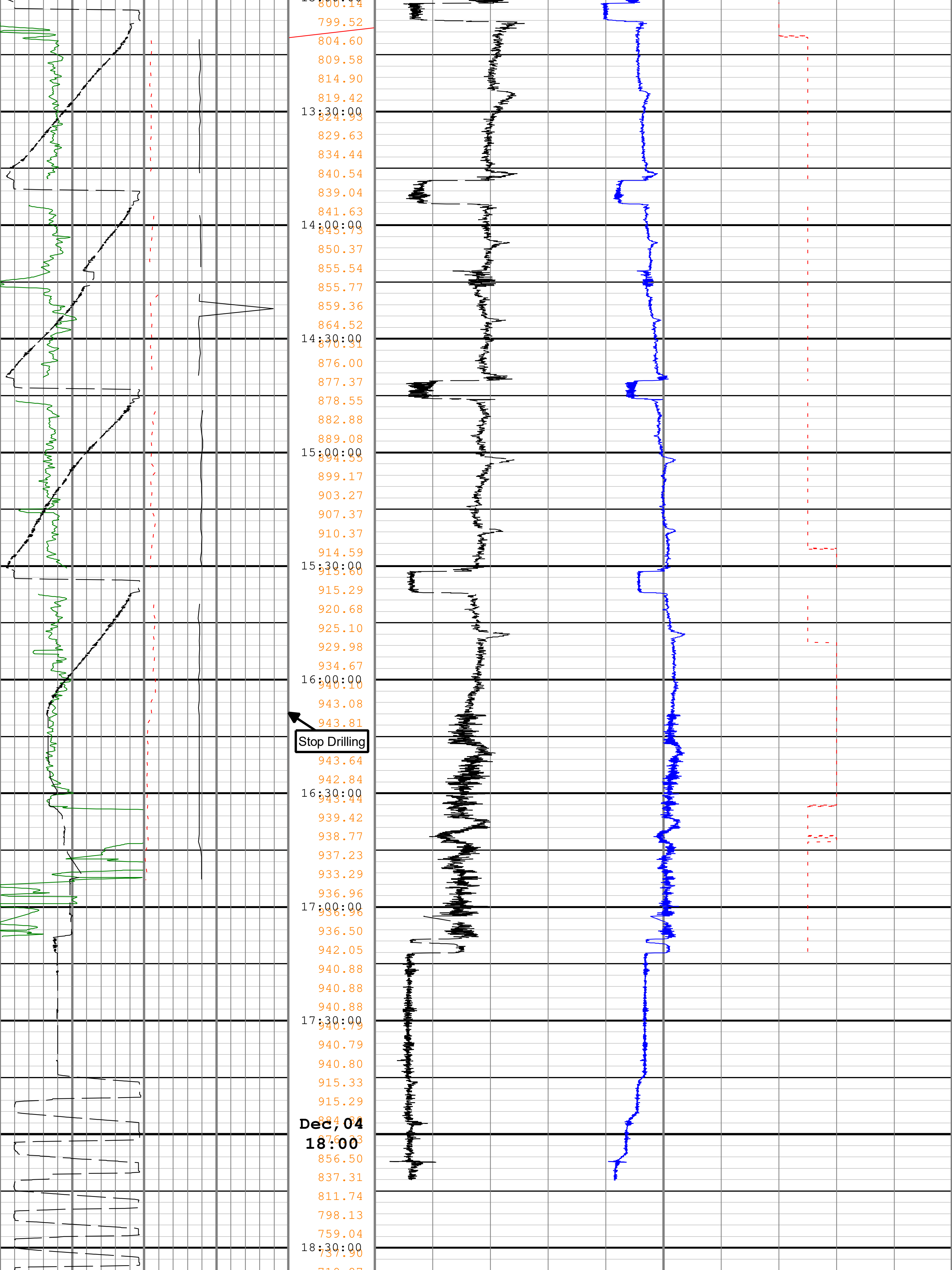
Track 1	Track 2	Track 3	Track 4
Gamma Ray (GRMA_ECO_RT) DV6MT RT 0 gAPI 150	Stick Slip Indicator (STICK_RT) TELE675 RT 0 c/min 500	Depth Index (DEPTH) RT m	Equivalent Circulating Density (ECD) DV6MT RM 1 g/cm3 2
Height of block above rig floor (BPOS) RT 0 m 45	Collar Rotational Speed (CRPM_RT) TELE675 RT 0 c/min 250	Shock Risk (SHKRSK_RT) TELE675 RT 5 0	Downhole Annulus Temperature (DHAT) DV6MT RM 0 degC 20
		Shock Peak (SHKPK_RT) TELE675 RT 0 gn 250	Downhole Annulus Pressure (DHAP) DV6MT RM 0 bar 200




420.79
424.00
02:00:00
431.32
435.35
439.29
443.10
446.86
02:30:00
451.34
456.40
456.30
461.03
464.95
03:00:00
476.44
481.93
487.31
492.81
492.43
03:30:00
505.31
511.56
514.18
520.84
523.72
04:00:00
532.19
532.78
539.12
537.96
540.88
04:30:00
539.07
536.76
536.92
541.23
542.83
05:00:00
552.56
556.88
561.08
565.83
569.90
05:30:00
573.46
577.57
579.41
581.55
Dec, 04
06:00
581.64
581.64
581.64
581.64
584.79
06:30:00
595.08
600.44
605.55
610.16
608.14
07:00:00
613.93
618.67
623.41
628.90



BPOS
CRPM_RT
SHKPK_RT
ECD
DHAP
DHAT
633.56
07:30:00
638.96
638.85
638.76
638.76
638.50
08:00:00
638.50
638.23
640.91
644.72
08:30:00
646.26
646.85
651.23
655.65
660.89
09:00:00
669.90
675.43
679.95
683.84
684.98
09:30:00
689.75
694.37
699.29
703.83
709.14
10:00:00
717.37
718.74
722.85
723.46
724.61
10:30:00
732.87
737.84
742.34
747.40
752.01
11:00:00
761.76
761.82
764.09
765.95
770.14
11:30:00
776.42
776.51
775.77
775.77
Dec, 04
12:00
775.77
775.77
775.77
775.77
775.77
12:30:00
780.43
786.07
791.40
796.09
800.18
13:00:00


799.52
804.60
809.58
814.90
819.42
13:30:00
824.93
829.63
834.44
840.54
839.04
841.63
14:00:00
845.73
850.37
855.54
855.77
859.36
864.52
14:30:00
870.31
876.00
877.37
878.55
882.88
889.08
15:00:00
894.55
899.17
903.27
907.37
910.37
914.59
15:30:00
915.60
915.29
920.68
925.10
929.98
934.67
16:00:00
940.10
943.08
943.81
Stop Drilling
943.64
942.84
16:30:00
943.44
939.42
938.77
937.23
933.29
936.96
17:00:00
936.96
936.50
942.05
940.88
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940.88
17:30:00
940.79
940.79
940.80
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Dec, 04
18:00
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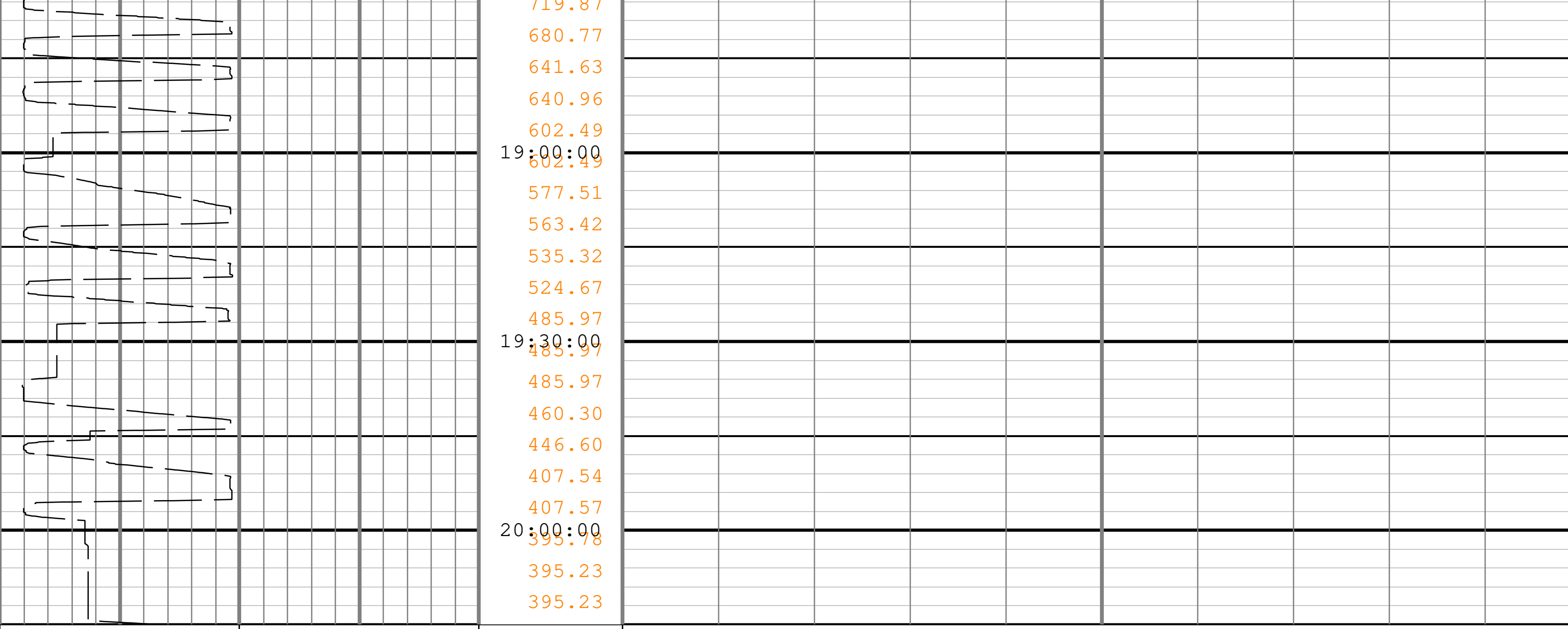
811.74
798.13
759.04
18:30:00
737.90



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Description: Format: Log (Drilling Mechanics Light) Index Scale: 5 cm per 3600 s Index Type: Time Creation Date: 11-Dec-2019 15:17:16

Company: Equinor

Well: 31/5-7

Field:	Eos
Rig Name:	West Hercules
Country:	Norway

Schlumberger

Drilling Mechanics - APWD

Time Based

Real Time, Recorded Mode